

From Web Evidence to Compact Verifiers: Distilling Verdicts and Rationales for Short-Video Misinformation Detection

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Abstract—A genuine short video can spread misinformation when its caption changes the event, place, or time. Fact-checkers therefore need both a verdict and a reason connecting the video’s claim to external evidence. This task is difficult because decisive claims may appear only on screen, retrieved information must address the same event, and a compact model must learn a short decision together with a longer explanation. Existing video methods commonly distill a classifier or obtain explanations from online large-model reasoning, leaving the final generative response outside the compact predictor. We introduce WEAVE, Web Evidence-Assisted Video Examination, to transfer this response task to a small language model. Its claim-centered interface separates textual claims, video accounts, model background, and Web context. A large reasoning teacher produces offline verdict–rationale targets; label-assisted selection and token-role-weighted supervision train the compact verifier to generate both outputs in one structured response. Large-model evidence preparation remains online, while teacher reasoning and prompt optimization remain offline. Evaluation on material from the FakeSV and FakeTT benchmarks examines teacher configurations, held-out teacher-correct examples, teacher-error examples, and student backbones. The one-billion-parameter verifier achieves strong, class-balanced retention, corrects a substantial share of teacher errors, and outperforms the evaluated two-billion-parameter configuration on both shared conditions. Case analysis connects its decisions to visual claim recovery and event-specific comparison. WEAVE makes prepared video and Web evidence actionable through compact verdicts and inspectable reasons, supporting model-assisted review of circulating clips.

Index Terms—short-video misinformation, Web evidence, multi-modal verification, knowledge distillation, compact language models

I. INTRODUCTION

A short video can be visually authentic and still communicate a false claim. This matters when a newsroom screens footage of an unfolding event, a platform reviews a widely shared clip, or a fact-checker revisits an old rumor. Recognizing the depicted objects or detecting editing artifacts cannot establish whether the caption describes the same event. Verification must recover the claim, identify the relevant event facts, and explain the relationship between them. The FakeSV [1] and FakeTT [2] benchmarks make this problem concrete through Chinese and English short-video misinformation, respectively.

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Real cases illustrate the information a verifier needs. Reuters traced footage presented as activity related to the 2023 Turkey earthquake to a 2018 SpaceX launch in California [3]. The images themselves did not establish the claimed event, location, or date. Another widely circulated rumor asserted that Minecraft would shut down in December 2020; reporting traced the announcement to a prank-news site [4]. In the screen-recording example analyzed here, the substantive announcement appears visually while the transcript contains only a closing phrase. A useful verifier must first find that assertion, then compare it with information about the announcement. These examples motivate preserving both visual claims and event-specific Web context.

Our research problem is to learn a *compact, evidence-conditioned verifier*: given a video’s textual and visual accounts and prepared background evidence, generate a real/fake verdict and a concise rationale stating the decisive agreement or discrepancy. Audiovisual interpretation and evidence preparation may use larger models, but the final verdict–rationale response should come from the compact model. This formulation serves review workflows in which the decision must be accompanied by a comparison that a person can inspect.

Three challenges shape this problem. **First, claim recovery is distributed across modalities.** A title, a spoken phrase, and visible text can carry different parts of an assertion. FakingRecipe models material selection and editing patterns [2], while NEED exploits neighboring videos [5]; these classification signals do not by themselves expose the precise claim that a generative verifier must assess. **Second, semantic similarity is weaker than event agreement.** A retrieved account may mention the right person but describe another action or date. NewsCLippings and COVE demonstrate the importance of contextual correspondence [6], [7]; Web-supported verification likewise requires evidence that addresses the claim [8]. A compact verifier needs these information roles to remain explicit after multimodal content is converted into text.

Third, compact classification and compact explanation generation are different learning targets. ExMRD supplies large-model reasoning to a small reviewer [9]. NVKE-CEI and CSI obtain explanatory analyses from online large-model checkers or agents before lightweight adjudication [10], [11].

Their small final predictors therefore do not perform the complete verdict–rationale generation task considered here. PCDD distills reasoning into a video detector [12], and Distilling Step-by-Step transfers label and rationale supervision to smaller text models [13]. These precedents establish the value of distillation; the remaining design question is how to connect prepared video/Web evidence to one compact response while giving its short verdict explicit learning priority.

We propose **WEAVE**, **Web Evidence-Assisted Video Examination**. Its central idea is to preserve information roles before transferring the final comparison task. A four-field interface separates the textual claim, video account, model-generated background, and Web-assisted context. Representative frames and a chronological description help retain visible assertions alongside speech, addressing claim recovery. Separating the two content fields from background knowledge then gives the student explicit material for event comparison. Web retrieval [14], [15] supplies external context rather than requiring the compact model to recall every relevant fact.

WEAVE next transfers a structured verdict–rationale response through offline supervision. A reasoning teacher uses a developed verification instruction to generate targets; label-assisted selection retains targets with correct verdicts. Token-role-weighted cross-entropy gives label tokens greater weight while retaining rationale supervision and normalizing over all active target tokens. The same verification instruction guides the teacher and student. This design aligns evidence organization, training targets, and inference: the small model learns both what decision to issue and how to express the comparison. Evidence preparation stays online; reasoning-teacher calls and prompt optimization stay offline.

We evaluate two instruction-tuned student backbones with nominal sizes of one billion (1B) and two billion (2B) parameters: Gemma-3-1B-it [16] and Gemma-2-2B-it [17]. Evaluation separates held-out teacher-correct examples from teacher-error examples to measure retention and correction. The 1B verifier performs strongly across class-balanced metrics in the first group, corrects a substantial share of the second, and exceeds the tested 2B configuration on both shared conditions. Case analysis inspects rationale support separately from label agreement [18], [19]. Our contributions are:

- **Claim-centered evidence organization.** Four explicit information roles make visually expressed claims and event context accessible to a compact text generator.
- **Compact joint response distillation.** Label-assisted selection and token-role weighting train one verifier to generate both the verdict and its reason from prepared evidence.
- **Actionable evidence about compact verification.** Teacher-conditioned evaluation reveals strong retention, useful error correction, and a favorable 1B–2B comparison; cases identify how claim recovery and event matching support inspectable decisions.

II. RELATED WORK

Video Misinformation and Context. The Fake Video Corpus [20], TikTec [21], and FANVM [22] establish complemen-

tary video-verification resources and approaches. FakeSV [1] and FakeTT [2] support short-video benchmarking, while NEED exploits neighboring content [5]. Transformers, CLIP, Whisper, and PP-OCR provide multimodal representation tools [23]–[26]. Visual News, NewsCLIPpings, and COVE emphasize contextual correspondence [6], [7], [27]. FakeSV-VLM adapts vision–language models for verification [28]; RADAR addresses topic shifts through retrieval-guided adaptation [29]; AdStyle targets writing-style robustness [30]. These approaches improve video representations, contextual matching, or robustness. WEAVE focuses on the complementary task of generating the final evidence-based response in a compact model.

Evidence Retrieval and Explanation. FEVER and FEVEROUS ground verification in evidence [31], [32]; AVeTeC uses real-world Web sources [8]. Retrieval-augmented generation [14], [15], ReAct [33], and Self-RAG [34] connect generation with retrieved information. ExMRD supplies staged reasoning to a reviewer [9]; NVKE-CEI combines content and evidence checkers [10]; CSI coordinates investigation and review [11]. Their explanations originate in the large-model analysis, and the final learned predictor performs classification. WEAVE instead learns the explanatory response inside the compact model. This distinction concerns where the comparison is generated, not simply whether an explanation exists. RARR, ALCE, FActScore, ARES, and Chain-of-Verification examine revision, citations, factual support, and verification [35]–[39]. Their distinctions inform our inspection of the relation asserted by a generated reason.

Compact Models and Response Supervision. Knowledge distillation [40] and chain-of-thought prompting [41] motivate transferring reasoning signals. Distilling Step-by-Step supervises label and rationale tasks [13]; PCDD transfers reasoning to a video detector [12]. LoRA and QLoRA enable parameter-efficient adaptation [42], [43], while DSPy and MIPROv2 develop instructions and demonstrations [44], [45]. Rationale evaluation distinguishes an accurate label from a supported explanation [18], [19].

Table I isolates three task differences: the evidence available to the final predictor, the location of online judgment, and the compact model’s output. ExMRD, NVKE-CEI, and CSI obtain large-model analyses before classification; PCDD transfers cognition into a detector. Distilling Step-by-Step demonstrates compact rationale learning on text tasks. WEAVE combines a video/Web evidence interface with joint verdict–rationale generation, moving the final explanatory response into the student while retaining large-model preparation. This addresses the need for both a compact decision and an inspectable comparison.

III. PROBLEM FORMULATION

A. Evidence and Outputs

Let $v_i = (b_i, u_i, t_i)$ denote example i , where b_i is its video-frame sequence, u_i its audio track, and t_i its title or description. The dataset label $y_i \in \{0, 1\}$ represents real content or fake/misleading content, respectively. The task is to predict

TABLE I

TASK AND OUTPUT COMPARISON. “LARGE” DENOTES AN ONLINE LARGE-MODEL REASONING STAGE; WEAVE RETAINS LARGE-MODEL EVIDENCE PREPARATION BUT USES THE STUDENT FOR FINAL JUDGMENT.

Method	Evidence / task	Online judgment	Compact output
ExMRD [9]	Video + knowledge	Large reasoning; reviewer	Label
PCDD [12]	Video perception + cognition	Distilled detector	Label
NVKE-CEI [10]	Video + Web	Large checkers; judge	Label
CSI [11]	Video case analyses	Large agents; adjudicator	Label
Step-by-Step [13]	Text tasks	Student	Label / rationale tasks
WEAVE	Four video/Web fields	Student	Joint verdict + rationale

this label and generate a concise rationale relating the video’s claim to the available information.

Let W_i denote the Web information accessible when the example is verified. An evidence-preparation function A maps the audiovisual input and this information to a text record:

$$\begin{aligned} x_i &= A(v_i; W_i), \\ x_i &= (R_{c,i}, R_{v,i}, K_{\text{int},i}, K_{\text{ext},i}). \end{aligned} \quad (1)$$

The fields are the reconstructed textual claim $R_{c,i}$, video account $R_{v,i}$, model-generated background $K_{\text{int},i}$, and search-assisted external knowledge $K_{\text{ext},i}$. Each field is text; we omit the example index when discussing its role. Search queries and source links are stored as provenance metadata outside x_i .

Given parameters θ and a verification instruction P^* , the compact student S_θ generates

$$(\hat{y}_i, \hat{e}_i) = S_\theta(x_i; P^*), \quad (2)$$

where $\hat{y}_i \in \{0, 1\}$ is its verdict and $\hat{e}_i$ is a natural-language rationale. WEAVE names both the framework and its compact verifier; backbone names identify the evaluated student configurations. The student receives the four text fields and performs the final comparison.

B. Claim–Evidence Comparison

Genuine footage can be paired with a false caption when the depicted event differs from the event asserted by the caption. A useful rationale therefore identifies the assertion, the relevant observation or event fact, and the agreement or discrepancy between them. Merely naming the same person or repeating an event description does not resolve a claim about what happened, where, or when.

Classification is evaluated against dataset labels. Evidence support is inspected by comparing the rationale’s stated relationship with the supplied record. This separates a correct verdict from the explanation offered for it. Labels are used for offline supervision selection and evaluation; the inference input contains the evidence fields and instruction. Compactness refers to the final generative verifier.

IV. WEAVE: EVIDENCE-CONDITIONED DISTILLATION

Figure 1 organizes WEAVE into seven components. Online, *content recovery* combines the title, speech transcript, representative frames, and chronological visual description. *Evidence preparation* uses a search-enabled large model to reconstruct the content and obtain event knowledge in one remote request. The *four-field interface* separates textual claims, video accounts, model background, and Web context. The *compact verifier* reads these fields with the shared instruction and generates the verdict–rationale response. Offline, *prompt development* optimizes the teacher’s reasoning instruction after evidence preparation. *Target construction* generates teacher responses and selects teacher-correct labels for the training pool. Finally, *student adaptation* learns QLoRA adapters using joint response supervision with label-prioritized token weights. The same instruction is retained for student training and inference. The preparation model and reasoning teacher have distinct roles even when they share a backbone; only preparation remains on the online path.

A. Multimodal Content Representation

WEAVE accepts a video and its accompanying title or description. For FakeTT, the description serves as the title field. Whisper Large V3 [25] transcribes the audio. If the video has no usable audio or transcription fails, the transcript is an empty string; the other inputs continue through the pipeline.

Two visual views serve different purposes. A representative-frame view supplies up to 16 images distributed over the clip. For a video of duration $d > 0$, the target time for frame j is

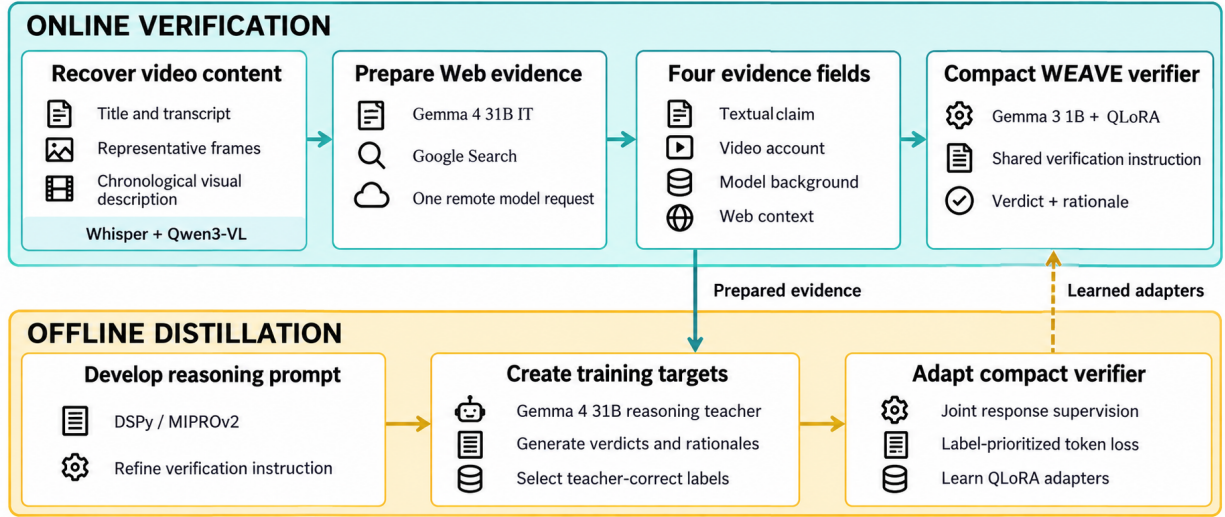
$$\tau_j = \frac{j}{15}d, \quad j \in \{0, \dots, 15\}. \quad (3)$$

The final target uses the last decodable frame near the end of the clip; unsuccessful reads are skipped. This view gives the preparation model direct access to visual details across the video’s duration.

A second view uses Qwen3-VL-4B-Instruct [46] to produce a chronological visual description. The implementation uses five-frame-per-second sampling with a cap of 120 frames and dynamic image resolution. The model runs with four-bit NormalFloat (NF4) weights on an RTX 5070 Laptop GPU with 8 GB of memory. Its prompt is: “Objectively describe the visual content of the video in temporal order. Respond in Traditional Chinese.” The chronological description supplies an event sequence that the preparation model can compare with the representative images. Representative images preserve on-screen assertions, and the chronological description organizes events. Both feed the video account, preserving visual claims even with little substantive speech.

B. Web-Assisted Evidence Construction

The preparation function A uses Gemma 4 31B IT [47] through the Gemini API, with temperature zero and Google Search enabled. A single external model request constructs the fields in Table II. The instruction organizes the title and transcript into R_c , then uses R_c , the visual description,



Large-model evidence preparation stays online. Teacher supervision and prompt development stay offline.

Fig. 1. WEAVE separates online evidence preparation and compact response generation from offline prompt development and distillation. The upper path recovers video content, prepares Web-assisted context, organizes four evidence fields, and generates a verdict with a rationale. The lower path develops the shared verification instruction, creates label-selected teacher targets, and learns the student adapters. The dashed arrow transfers the learned adapters into the online verifier.

TABLE II

THE EVIDENCE INTERFACE. EACH FIELD HAS AN EXPLICIT INFORMATION ROLE, AND ALL FOUR FIELDS ENTER THE STUDENT.

Field	Construction	Decision role
R_c	Title and transcript	Recover textual assertions.
R_v	Claim, visual description, and frames	Locate visual claims and event observations.
K_{int}	Parametric model background	Supply related entity and event knowledge.
K_{ext}	Search-assisted summary	Check the claim against external accounts.

and sampled images to construct R_v . It distinguishes model-generated background K_{int} from information obtained through Web search, K_{ext} .

Table III specifies concise prompt contracts for preparation and verification. The fields share a generation context, with R_v informed by R_c ; their names distinguish each source and decision role. Search queries use entities, actions, locations, and dates to target the assertion and relevant background. The instruction favors authoritative sources and summarizes verification-relevant content in K_{ext} . Deduplicated queries and source links remain metadata for inspecting the summary’s provenance.

C. Teacher Responses and Prompt Development

The large-model teacher uses Gemma 4 31B IT [47] with temperature zero to generate response supervision. Let T denote this teacher and P^* the finalized verification instruction. For evidence record x_i ,

$$(\tilde{y}_i, \tilde{e}_i) = T(x_i; P^*), \quad (4)$$

where $\tilde{y}_i \in \{0, 1\}$ is its label and $\tilde{e}_i$ its generated rationale. These two outputs form the student’s response target. The student JSON uses the keys `pred_label` and `reason`, preserving the same label meanings at training and inference.

Prompt development applies specifically to the reasoning stage. DSPy [44] and MIPROv2 [45] optimize the verification instruction and demonstrations consumed by the large-model teacher after the four evidence fields have been prepared. The development score places weight 0.95 on one minus the squared difference between the gold label and predicted fake probability. It adds 0.05 when the stripped reason contains at least 50 characters. This score guides instruction search; the final response contract uses a binary label. Qualitative error analysis then informs the final task-specific instruction. The six comparison checks in Table III specify verification questions, with the output remaining binary.

The resulting instruction P^* first guides the large-model teacher in generating offline verdict–rationale targets and is retained as the verification instruction for student training and inference. DSPy and MIPROv2 operate only during prompt development. At inference, the evidence-preparation function A supplies the four fields and the trained student generates the final response, with no call to the reasoning teacher T or prompt optimizer.

D. Verdict–Rationale Distillation

WEAVE uses teacher-response supervision rather than matching teacher logits. Let $\mathcal{I}$ be the index set of aligned evidence records, labels, and teacher responses. Define the teacher-correct pool

$$\mathcal{C} = \{i \in \mathcal{I} : \tilde{y}_i = y_i\}. \quad (5)$$

Class-stratified partitioning of $\mathcal{C}$ gives disjoint training, validation, and held-out subsets $\mathcal{D}_{\text{tr}}$, $\mathcal{D}_{\text{val}}$, and $\mathcal{E}_1$. The teacher-error

TABLE III

PROMPT CONTRACTS FOR EVIDENCE PREPARATION AND FINAL VERIFICATION, EXPRESSED IN ENGLISH. THE PREPARATION CONTRACT SPECIFIES SOURCE ROLES; THE VERIFICATION CONTRACT SPECIFIES THE COMPARISON AND JSON OUTPUT.

Component	Instruction content
Preparation role	Analyze the video objectively. Preserve the people, events, places, and times needed for fact checking. Use the supplied observations and the specified knowledge source for each field. Return plain JSON with keys <code>Rc</code> , <code>Rv</code> , <code>K_int</code> , and <code>K_ext</code> ; omit Markdown fences.
Source constraints	Construct <code>Rc</code> from the title and transcript. Construct <code>Rv</code> from <code>Rc</code> , the chronological visual description, and ordered frames; preserve visible assertions and event order. Use model knowledge for <code>K_int</code> . Use Web search for <code>K_ext</code> , targeting the specific event and relevant background. Keep model recollection and search-derived information in their respective fields.
Verification role	Act as a short-video fact checker. Assess the assertion conveyed by the textual and visual accounts against the internal and external background. Identify the claim, inspect the video account, compare the evidence, and produce a verdict with a concise reason.
Comparison checks	Check for reused footage with a changed context; reassigned locations or political settings; incorrect entities, dates, or other precise details; attachment of an unrelated narrative; physical implausibility; and contradictions between the textual claim and visible events.
Evidence rules	Refer to both knowledge fields. When Web context does not resolve the claim, examine the supplied observations and model background. Do not infer falsity solely from missing search support. Do not invent an absent day, month, or year and then use it to allege a contradiction.
Input / output	Supply the four fields under separate headings. Return one JSON object with <code>pred_label</code> and <code>reason</code> . Use integer 0 for real content and 1 for fake or misleading content. The reason should state the decisive agreement or discrepancy in clear, concise language.

group is $\mathcal{E}_2 = \{i \in \mathcal{I} : \tilde{y}_i \neq y_i\}$. Test 1 and Test 2 refer to $\mathcal{E}_1$ and $\mathcal{E}_2$. Stratification preserves class proportions. Selection aligns the supervised verdict with the dataset label, while the teacher’s explanation supplies the associated response text.

The principal student is Gemma-3-1B-it [16]; Gemma-2-2B-it [17] is a second backbone. For training example i , the instruction and four fields form a textual prefix h_i . The teacher’s label–reason response forms target sequence $z_i = (z_{i1}, \dots, z_{im_i})$ with m_i tokens. Let $\mathcal{T}_i^y$ and $\mathcal{T}_i^e$ be the nonempty, disjoint sets of supervised target positions associated with the label and rationale. Their union $\mathcal{T}_i = \mathcal{T}_i^y \cup \mathcal{T}_i^e$ contains all active target positions. For a finite set $\mathcal{B}$, $|\mathcal{B}|$ denotes its cardinality; thus $N_{\text{act}} = \sum_{i \in \mathcal{D}_{\text{tr}}} |\mathcal{T}_i|$ counts the active target tokens across the training set.

To prioritize verdict prediction while retaining rationale supervision, WEAVE minimizes token-level weighted cross-entropy normalized over these active tokens:

$$\mathcal{L}(\theta) = -\frac{1}{N_{\text{act}}} \sum_{i \in \mathcal{D}_{\text{tr}}} \sum_{t \in \mathcal{T}_i} w_{it} \log p_{\theta}(z_{it} \mid h_i, z_{i,<t}), \quad (6)$$

where p_{θ} is the student’s next-token distribution, $z_{i,<t}$ denotes the target tokens preceding position t , and the token-role weight is given by:

$$w_{it} = \begin{cases} \lambda_y, & t \in \mathcal{T}_i^y, \\ \lambda_e, & t \in \mathcal{T}_i^e, \end{cases} \quad (\lambda_y, \lambda_e) = (5.0, 1.0). \quad (7)$$

Each label token receives five times the weight of a rationale token. The common denominator averages over active output positions, while the role weights emphasize verdict prediction within the joint response. Rationale tokens continue to supervise the expression of the evidence comparison. All supervised positions update the same student adapters.

Prompt and padding positions are masked from the loss, and dynamic padding supplies an attention mask. Training uses QLoRA [43]: quantized base weights remain frozen while low-rank adapters [42] are optimized. The adapters cover query, key, value, output, gate, up, and down projections. The 1B adapter configuration uses rank 16 and LoRA alpha 32, giving the standard adapter scaling factor $32/16 = 2$. Table IV summarizes adaptation and decoding settings. The checkpoint with the lowest validation loss is selected.

At inference, the instruction and four fields are formatted with the backbone’s chat template. Greedy generation produces the response, and only newly generated tokens are decoded. The parser reads the JSON object and uses named-field extraction when surrounding formatting interferes with parsing. This keeps the final interface a verdict and a reason, without adding a separate classifier head.

V. EXPERIMENTAL SETUP

A. Data and Evaluation Protocol

Table V summarizes the source collections. FakeSV [1] contributes Chinese short videos, and FakeTT [2] contributes TikTok material. The usable inventory contains labeled examples successfully processed by the large-model pipeline. Examples blocked by model response policies or other processing constraints are excluded. FakeSV is nearly balanced, whereas FakeTT contains more fake examples. Macro metrics complement accuracy by assigning equal importance to each class.

The student evaluation has two groups defined by the teacher’s original prediction. *Test 1* contains 507 held-out examples from the teacher-correct pool. *Test 2* contains 802 teacher-error examples withheld from student training. Their

TABLE IV

STUDENT ADAPTATION AND EXECUTION. PARAMETER COUNT REFERS TO THE 1B ADAPTER CHECKPOINT, MEMORY TO TRAINING, AND GENERATION TIME TO THE LOCAL STUDENT.

Item	Setting or observation
Base quantization	4-bit NF4; double quantization
Adapter rank / alpha / dropout	16 / 32 / 0.05
1B adapter parameters	13,045,760
Epochs / microbatch	5 / 1 example
Gradient accumulation	4 steps
Optimizer / learning rate	AdamW / 2×10^{-4}
Weight decay / warmup	0.01 / 20 steps
Maximum gradient norm	1.0
Training sequence length	1,024 tokens
Precision	Mixed precision; bf16 when supported
Checkpoint selection	Minimum validation loss
Training peak memory (1B)	3.88 GB
Decoding	Greedy; maximum 256 new tokens
Student generation time	Approximately 12–15 s per example
Evaluation wall time	Approximately 4–5 h for over 1,300 examples

TABLE V

SOURCE COLLECTION INVENTORY. REAL/FAKE COUNTS REFER TO USABLE VIDEOS; DATE RANGES REFER TO VIDEO PUBLICATION.

Collection	Original	Usable	Real	Fake
FakeSV	3,624	3,609	1,802	1,807
FakeTT	1,991	1,990	819	1,171
Total	5,615	5,599	2,621	2,978

FakeSV: October 2017–February 2022. FakeTT: May 2019–March 2024.

union contains 1,309 examples and is called the *pooled diagnostic set*. The groups evaluate retention on teacher-correct examples and correction on teacher-error examples. The pooled set combines these two conditions in their observed proportions. Web retrieval occurs at verification time and can include later reporting or existing fact-checks; the task is retrospective verification of circulating videos.

The teacher development study compares three configurations: an ExMRD-style direct large-model decision without its Small Language Reviewer (SLR), WEAVE’s initial verifier, and the developed verifier after DSPy-assisted instruction search and manual refinement. Each configuration is summarized on its evaluation collection. These development measurements characterize the supervision-producing configurations; the student analysis uses the two explicitly defined groups above.

B. Metrics and Computational Measurements

Student evaluation uses accuracy and macro-averaged precision, recall, and F1. For class $c \in \{0, 1\}$, let TP_c , FP_c , and FN_c count true positives, false positives, and false negatives

TABLE VI

ACCURACY OF TEACHER DEVELOPMENT CONFIGURATIONS ON THEIR RESPECTIVE EVALUATION COLLECTIONS. THE TABLE DESCRIBES CONFIGURATION-LEVEL BEHAVIOR; STUDENT EVALUATION IS SHOWN SEPARATELY IN FIG. 2.

Configuration	Accuracy (%)
ExMRD-style direct decision, no SLR	68.36
WEAVE, initial verification prompt	73.23
WEAVE, developed verification prompt	86.39

when c is the positive class. The class-specific scores are

$$\begin{aligned} \text{Prec}_c &= \frac{TP_c}{TP_c + FP_c}, & \text{Rec}_c &= \frac{TP_c}{TP_c + FN_c}, \\ \text{F1}_c &= \frac{2TP_c}{2TP_c + FP_c + FN_c}. \end{aligned} \quad (8)$$

For $M \in \{\text{Prec}, \text{Rec}, \text{F1}\}$, the macro average is $M_{\text{macro}} = (M_0 + M_1)/2$. Accuracy is the fraction of examples with $\hat{y}_i = y_i$. Macro-F1 thus averages class-specific F1 values. A class score with a zero denominator is defined as zero.

Computational measurements distinguish external-model usage from local student execution. The external profile measures model requests, mean token use, and mean API processing time per video. Requests exclude local speech recognition, visual description, and student inference; service-internal searches are part of the search-enabled request. The ExMRD-with-SLR reference uses its remote refinement, retrieval, and reasoning stages, whereas WEAVE’s student pipeline uses the remote preparation request. Table IV separately characterizes adapter training and local response generation.

VI. RESULTS AND ANALYSIS

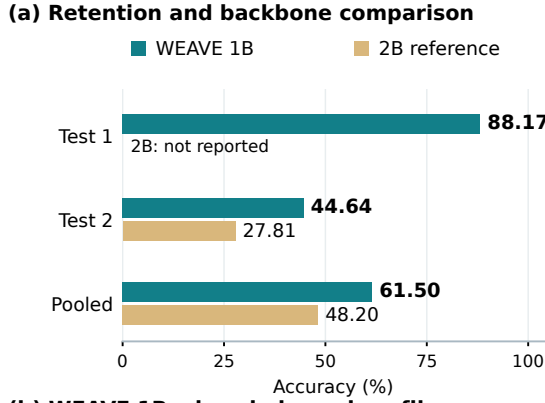
A. RQ1: Teacher Development Configurations

The developed WEAVE verifier achieves the highest accuracy among the measured teacher configurations in Table VI; the initial configuration also exceeds the ExMRD-style direct decision. These scores identify the developed verifier as the strongest supervision-producing configuration in this study. They summarize complete configurations on their respective evaluation collections.

The developed instruction translates verification into concrete comparisons: identify the assertion, locate the relevant event information, and state the agreement or conflict. It distinguishes a match in entity identity from a match in the claimed location, quotation, or event, and treats missing search support separately from evidence of falsity. This task definition gives the teacher an explicit comparison to perform and produces supervision aligned with the student’s final response. Its practical benefit is a consistent evidence-to-verdict interface throughout prompt development, target generation, and compact inference.

B. RQ2: Teacher-Conditioned Student Performance

Figure 2 shows strong retention by the 1B verifier. On held-out teacher-correct examples, accuracy and all three metrics are consistently high and closely aligned. The compact model



(b) WEAVE 1B: class-balanced profile

	Precision	Recall	F1
Test 1 n = 507	88.99	88.09	88.09
Test 2 n = 802	36.30	35.58	35.91
Pooled n = 1,309	58.81	58.73	58.76

All scores in %; darker cells indicate higher scores.

Fig. 2. Student performance (%). (a) Accuracy of the 1B verifier and the 2B reference; only shared conditions support backbone comparison. (b) The complete 1B macro-metric profile. Test 1 measures held-out teacher-correct retention, Test 2 measures teacher-error correction, and Pooled combines them. Sample counts refer to the 1B evaluation.

therefore preserves useful teacher behavior across both classes. This is a central benefit of the framework: the evidence fields and offline response targets support a small generator that performs the final verification task and explains its decision. On Test 2, the student correctly resolves a substantial share of examples that the teacher misclassifies. These teacher-error responses never supply student training targets, so successful correction shows a learned decision mapping that goes beyond reproducing the teacher’s original labels. The lower conditional scores identify teacher-error examples as the harder verification group. Viewed together, the two groups reveal the useful combination delivered by WEAVE: strong retention of correct decisions and additional corrections where the teacher fails.

The pooled profile lies between the two groups. This aggregate has an exact interpretation that is useful when comparing evaluation designs. Let $n_1 = |\mathcal{E}_1|$ and $n_2 = |\mathcal{E}_2|$ be the group sizes, Acc_1 and Acc_2 the student accuracies within them, and $\pi = n_1/(n_1 + n_2)$ the fraction of Test 1 examples. Accuracy on the union is: $\text{Acc}_{\text{pool}} = \pi \text{Acc}_1 + (1 - \pi) \text{Acc}_2$. Pooled accuracy depends on both group proportions and conditional accuracies. Macro-F1 uses pooled class-specific confusion counts and generally differs from this weighted average. Conditional profiles therefore preserve the distinction between retention and correction.

C. Student Backbone Comparison

The adapted Gemma-3-1B-it verifier outperforms Gemma-2-2B-it in both the teacher-error and pooled conditions in

Remote computation per video

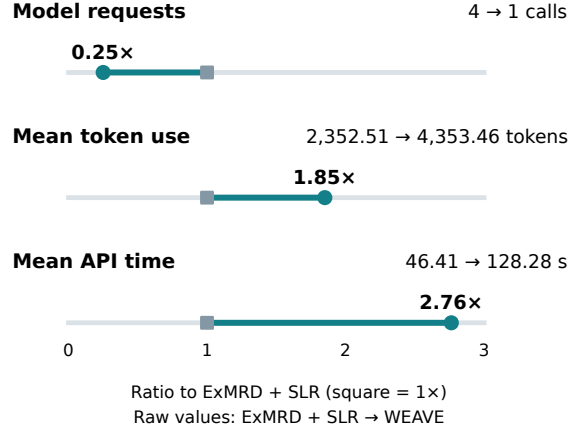


Fig. 3. External-model workload. Squares mark the ExMRD-with-SLR reference and circles show WEAVE on a common ratio scale. Each row gives both raw values. The one-call preparation stage reduces remote invocations while using more tokens and API time. Local video processing and student decoding are measured separately.

Fig. 2. This consistent ordering makes the 1B configuration the stronger choice among the evaluated compact verifiers. This is useful for model selection: parameter count alone does not determine the quality of evidence-to-response generation, and the tested configurations also differ in pretrained model family. Preserving claim details, comparing events, and following the response contract depend on pretraining and adaptation as well as model size. The 1B configuration combines the stronger measured decision profile with the smaller nominal backbone and supplies the responses examined below.

D. Computational Characteristics

Figure 3 shows how WEAVE reallocates online computation. It consolidates remote work into one evidence-preparation invocation and delegates verdict–rationale generation to the local student. Its mean remote token use and API processing time are higher than the reference, reflecting more work within that preparation request. The aligned rows and common ratio scale distinguish invocation count from tokens and time.

The single remote request performs content reconstruction, background generation, and Web-assisted context preparation. The architectural benefit is a clear dependency boundary: once the four fields are available, the student produces the complete response locally. Teacher target generation and prompt optimization remain offline. This separation makes the evidence preparation and final verifier independently identifiable computational stages.

Table IV characterizes local execution through adapter size, training peak memory, response-generation time, and evaluation wall time. A frozen quantized backbone with learned adapters generates both the label and reason. These measurements establish a concrete resource profile for a compact, autoregressive verifier whose output contains an inspectable explanation as well as a decision.

From visual claim to supported verdict

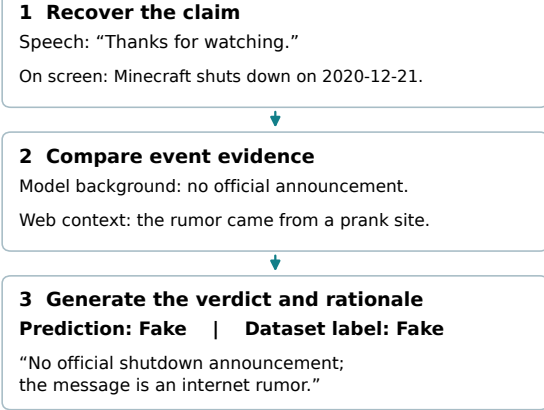


Fig. 4. A visually expressed shutdown rumor. The video account recovers the assertion absent from speech, event evidence rebuts it, and the compact verifier expresses the resulting decision. Field contents and the student reason are condensed into English. The rumor’s prank origin is documented by Prima Games [4].

E. RQ3: Claim–Evidence Relationships in Generated Reasons

Figure 4 traces the Minecraft example from content recovery to the final response. The transcript supplies only a closing phrase. The video account recovers the substantive assertion: a displayed announcement that Minecraft will shut down on a specified date. External context traces the announcement to a prank, and the student’s reason identifies the absence of an official shutdown announcement. The useful comparison concerns the claimed event, rather than incidental details of the phone interface.

Table VII adds two event-based examples. In the DeSantis case, the speech concerns a border deal, and the visual account identifies the speaker and West Des Moines setting. External reporting describes the same remarks. The student’s reason points to this agreement, supporting its real verdict through event content as well as identity. The Fallon–Trump record illustrates why the reason is an independently useful output. Although the prediction matches the dataset label, the reason alleges a mismatch with footage of the same interaction. That stated mismatch is not established by the displayed record. Reading the reason therefore directs attention to the precise comparison that needs inspection, rather than treating label agreement as evidence that every supporting statement follows.

The cases define three useful inspection criteria: rebuttals should address the same assertion, event agreement should match actions or remarks, and contradictions should identify conflicting content. The four-field interface makes these comparisons accessible.

VII. DISCUSSION

A. What the Results Imply for Compact Verification

The results establish that a compact generator can retain strong teacher-correct performance, correct teacher errors, and express the comparison behind its verdict. Five design insights

explain how WEAVE organizes this task and why its observed behavior is useful.

- **Preserve the claim wherever it appears.** The Minecraft visual account supplies the assertion absent from speech. The student uses it without processing raw frames. The practical value is claim coverage: preserve the entity, action, and date that make a statement checkable, even when another modality contributes little.
- **Give the student a focused comparison task.** In the DeSantis case, the prepared evidence connects the speaker, location, and specific border-deal remarks. The student can state a concrete event agreement from the text fields, without interpreting raw video or issuing a search. This division of labor explains the practical value of the interface: the compact verifier receives the claim and comparison material in a consistent form.
- **Align the verdict target and emphasize its tokens.** Teacher-correct target selection aligns the supervised verdict with the dataset label. Equations 6–7 then give each label token greater weight within the joint response, while rationale tokens supervise the expression of the comparison. The active-token denominator excludes prompt and padding positions. Together, these choices give a short decision explicit priority while preserving explanation supervision in the same model.
- **Select backbones using both retention and correction.** The 1B verifier exceeds the 2B configuration on both shared conditions. Its strong teacher-correct profile and successful teacher-error corrections show why the two groups matter: one measures preservation of useful supervision, and the other identifies decisions that improve on the teacher’s original predictions. Their combination supports choosing compact verifiers by demonstrated task behavior.
- **Inspect the relation asserted by a reason.** The Fallon–Trump case separates label agreement from rationale support. Identifying both sides of the comparison lets a reader test the alleged agreement or discrepancy against the evidence. This makes rationale generation directly useful for model-assisted review.

B. Use in Evidence-Based Review

WEAVE supports retrospective review of circulating clips by presenting reconstructed content and event evidence alongside the verdict. In a newsroom, an editor checking a recycled clip can read the recovered assertion and compare the named event with the retrieved account. This is the distinction illustrated by the launch footage in the Introduction: an authentic image needs an event-level explanation before it can support a caption. In platform review, a visually displayed announcement can be checked even when the audio contributes no substantive claim. The Minecraft case shows the value of carrying that announcement into the student’s input. For a political quotation, the useful reason identifies the reported action or remarks, as in the DeSantis example, rather than relying only on recognizing the speaker. These scenarios share the same output requirement: a decision accompanied by a short, inspectable comparison. WEAVE meets that requirement through one compact generator

TABLE VII
EVENT-BASED CASE COMPARISON. CONTENT AND REASONS ARE CONDENSED INTO ENGLISH. THE FINAL COLUMN IDENTIFIES THE EVIDENCE RELATIONSHIP A READER CAN INSPECT; DATASET-LABEL AGREEMENT IS SHOWN SEPARATELY.

Case	Video account and claim context	Retrieved event context	Prediction / dataset label	Student reason and inspection target
DeSantis speech	C-SPAN footage in West Des Moines; remarks criticize Republican senators’ handling of a border deal.	Reporting identifies the speech and the same criticism of the border deal.	Real / Real	Says the visual content and event background are consistent. Inspect the matching action and remarks.
Fallon–Trump clip	A television-studio interaction shows Fallon ruffling Trump’s hair; speech consists of exclamations.	Retrieved accounts identify the same interview and interaction.	Fake / Fake	Alleges inconsistency with the interview footage while describing the same interaction. Inspect whether any specific discrepancy is established.

with a consistent response contract. Reviewers can move from the reason to the relevant evidence field, and then to source metadata when the external account warrants closer reading. The practical benefit is continuity between the decision, its stated basis, and the material used to make it.

Separate computational measurements guide optimization of the two stages. Request count describes remote dependencies, tokens and API time describe evidence preparation, and student storage and decoding determine local requirements. This distinction helps match the verification workflow to available resources.

C. Evidence Granularity and Further Development

Claim-focused selection can preserve decisive entities, actions, and temporal qualifiers within a fixed token budget; long-context research [48] motivates studying evidence placement and use. Passage-linked supervision can associate rationale statements with supporting sources [36]–[38]. Source- and event-separated evaluation can characterize changes in topic and publication context. Verification-oriented generation [35], [39] also motivates selecting targets by claim coverage and comparison consistency. Temporal evidence policies can separate retrospective review from assessment using only information available at a specified time. An evidence-insufficient outcome would broaden the decision space. These extensions preserve the compact response interface while strengthening traceability to evidence.

VIII. CONCLUSION

WEAVE distills verdicts and rationales into a compact verifier through four fields for textual claims, visual accounts, model background, and Web context. Offline teacher targets and token-role-weighted supervision train one structured response generator. The 1B verifier demonstrates strong, class-balanced retention on teacher-correct examples, corrects a substantial share of teacher errors, and outperforms the evaluated 2B configuration on both shared conditions. Visual claim recovery and event matching make the evidence actionable, while generated reasons expose the student’s comparison for inspection.

WEAVE retains large-model evidence preparation and delegates the final response to the compact student. Future work

will pursue claim-focused selection, passage-linked rationale supervision, temporal and source-separated evaluation, and evidence-insufficient decisions, strengthening the connection between predictions and supporting information.

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